

Yicheng MA
(+65) 80153972 | YICHENG003@e.ntu.edu.sg
222 WESTWOOD AVENUE #10-11 SINGAPORE 648355

EDUCATION

Nanyang Technological University	09/2024 – Present
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Master of Science (MSC) in Signal Processing and Machine Learning

GPA: 4.25/5

Courses:

Machine Vision, Artificial Intelligence & Data Mining, Genetic Algorithms & Machine Learning, Probability & Random Processes

Zhejiang University	09/2019 – 06/2024
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Bachelor of Science in Engineering (BSE) in Opto-Electronics Information Science and Engineering

GPA: 3.92/4.0 (Ranking: Top 15%)

Courses:

Probability Theory and Mathematical Statistics, Signal and Processing, Machine Vision and Image Processing, Embedded Systems and Applications etc.

ENTREPRENEURSHIP EXPERIENCE

Gstarcad Internet	Hangzhou, China
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Role: CEO (The Establishment, Maintenance, and Operation of the Platform)

07/2022 – 07/2023

- Proposed the idea of building a virtual talent pool in the forms including Web pages, mini-program, APP, etc. to alleviate the problems in hometown, such as outsourcing programs and education resource integration
- Developed a team with members from top universities in China and established the company Gstarcad Internet
- Engaged in Web development and WeChat operations and initiated the project of *Call You Back SEIFO*
- Coordinated teammates in Web development and organized various thematic activities such as Chinese Valentine's Day—Couple Matchmaking that connected the Internet and traditional cultures

INTERNSHIP

ASTAR SIMTech ARM	Singapore
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Role: *Research Intern*

09/2024 – Present

- Developed a UR10e grasping system designed to operate in uncertain and dynamic clustered environments. This project involves exploring and leveraging RTDE (Real-Time Data Exchange) as a motion planning tool to enhance system reliability and responsiveness.
- Enhanced the functionality of AnyGrasp, a state-of-the-art grasp generator, by integrating the Grasp-1Billion dataset with MetaGraspNet. This integration requires preprocessing and aligning the datasets to ensure consistency in their structure and arrangement, enabling seamless application and improved grasping performance.
- Conducted simulations using NVIDIA Isaac Sim, replicating the grasping system in a simulated environment to evaluate its performance under various scenarios. Currently, I am exploring novel methods to transform task-agnostic grasp detectors into task-oriented systems. Potential approaches include leveraging Diffusion models or Reinforcement Learning (RL).
- Planning to submit research findings and contributions to IROS 2025, focusing on innovative techniques and their applications in robotic grasping systems.

Sunny Optical Technology Group	Hangzhou, China
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Role: *PA Engineer*

06/2022-07/2022

- Explored and studied the test principles of inverse optical path device, car lens and HUD system, and principle of tolerance and yield analysis
- Acquired the use of ZEMAX and AutoCAD and applied them to optical practice and designed a simplified camera optical imaging system
- Tested the parameters and performance index of multiple imaging lens and produced analytical reports

PROJECT RESEARCH

Robotic Manipulation System using Advanced Deep Learning Technique

09/2024 – Present

Research Intern of the ASTAR SIMTech ARM, Singapore

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- Planning to submit research findings and contributions to IROS 2025, focusing on innovative techniques and their applications in robotic grasping systems.

Graduation Project and Thesis

08/2023 – 06/2024

Topic: Research on static and dynamic grasping of robots for warehousing and logistics applications

- Constructed a geometric grasping module for grasp generation. This article refers to the GSNet model architecture, using graspness to measure points in the point cloud suitable for grasping and extracting local and global high-dimensional point cloud features. Then, it extracts point-wise grasping degrees and subsequent viewpoint-wise grasping degrees, sets a graspness threshold to filter target point clouds, and generates grasps based on the filtered target point clouds, achieving static grasp generation.
- Built a dynamic tracking module based on time-based graspness. This article refers to AnyGrasp, using multi-threaded high-dimensional feature vectors to represent each grasp in each frame, calculating cosine similarity to measure the similarity between high-dimensional feature vectors, and using this to measure the correspondence of temporal dimensions between grasps in frames, achieving continuous generation of dynamic tracking poses.
- Established a robot motion control and path planning system. This article, based on ROS (Robot Operating System), utilizes ROS's distributed communication architecture to connect and communicate between nodes. Motion control and path planning mainly use MoveIt API to write robot control modes and path planning, achieving multidimensional control of the robotic arm system.
- Designed static and dynamic experiments for robot new object grasping. Static experiments include parcel grasping experiments and daily necessities grasping experiments to verify the generalization ability of the constructed robot new object grasping system. Dynamic experiments involve grasping parcels moving on conveyor belts to verify their dynamic grasping capability. Through experimental verification, this article demonstrates that the robot learning-based unknown object grasping system constructed in this article exhibits good performance in both static and dynamic object grasping scenarios.

Microstructure Fiber Array Single Virus Sensing System Based on Microimaging

03/2022 – 07/2022

Leader of the Establishment of Single Molecule Fluorescence Microscope and Enumeration APP (Research Group)

- Produced a microimaging-based single virus sensing system that 1) used pores inherent in the porous microstructure fiber as the array for sample circulation 2) processed advanced microimaging like hyper-spectrum and enumeration on each pore on optical fiber face 3) measured the trace virus of ultra-low abundance to realize hypersensitivity and specificity 4) adjusted pore size and cycle to enhance flux detection
- Built a portable fluorescence microscope to replace the fixed lab-used large microscope, developed a simple and portable system for the lab bulky dPCR instrument, completed the enumeration programming for mobile APP and MATLAB
- Accomplished detection & analysis integrated operations by constructing a relatively cheap optical microscope and mobile APP, selected and used LED light sources with shorter wavelengths or narrower bandwidths on future experiments (with utility model patent and invention patent in progress)

Student Research Training Program (SRTP)

03/2021 – 04/2022

Topic: UVA Airborne Laser Radar Sounding Technology | Role: Mathematical Modeling and Simulation Leader

- Conducted complete simulation and reflection on scanning process of current airborne simulators including Shoals System, Wa-LiD simulator, etc. from the aspects of inertial navigation, GPS, precision, etc.
- Analyzed the sounding efficiency of the single signal of simulators and explored the noise analysis during scanning
- Applied MATLAB to develop the mathematical simulation model and completed the conceptual airborne laser radar through SolidWorks; simulated real situations of the scanning by embodying the functions including rotation, scanning, inversion, etc.

ACTIVITIES

Director of Academic Division, Student Union, ZJU College of Optical Science and Engineering 10/2021- 6/ 2022

- Led teams to make surveys on the needs of students and planned corresponding academic events ranging from seminars and contests to lectures to enrich their knowledge system
- Collaborated with other divisions and organizations to schedule academic and technology festivals to improve the intellectual atmosphere
- Launched experience exchange on the study of different courses and invited model students to share their methods and experience and bridge the communication between students and professors

Member, Network Publicity Department, Film Society of Zhejiang University

10/2019 - 06/2020

- Followed up latest movies online and cooperated with teammates to organize significant movie-watching and applied social media platforms to attract more audience and expand the influence of activities

Volunteer, The Second Affiliated Hospital Zhejiang University School of Medicine

10/2021

SKILLS AND OTHERS

Interested Area: **Robotics or related fields**

Computer Skills: Python, C, VSCode, Ubuntu, ROS, OpenCV, Open3D, Pytorch(deep learning), Isaac Sim, MATLAB, Github, etc.

Interests: Swimming, Cycling, Football, Reading